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```
clc;
clear;
close all;
```

PARAMETERS

```
M = 16;
k = log2(M);
Nbits = 4000;
```

GENERATE RANDOM BITS

```
txBits = randi([0 1], Nbits, 1);
```

16-QAM MODULATION

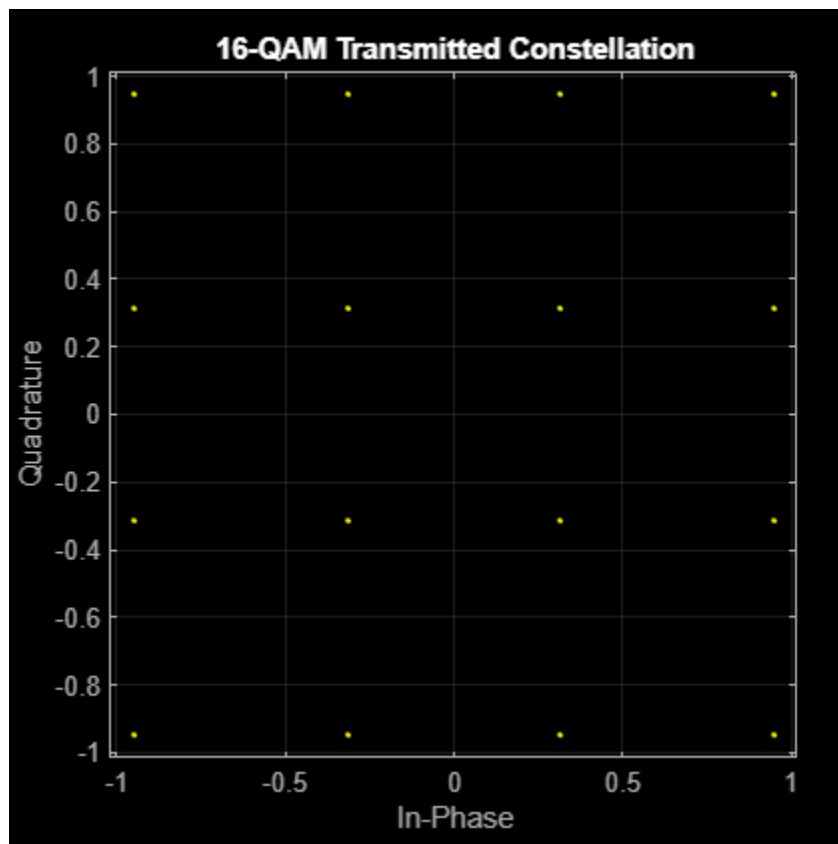
```
txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);
```

TRANSMITTED CONSTELLATION

```
figure;

scatterplot(txSymbols);

title('16-QAM Transmitted Constellation');
grid on;
```



16-QAM DEMODULATION

```
rxBits = qamdemod(txSymbols, M, ...  
    'OutputType', 'bit', ...  
    'UnitAveragePower', true);
```

BER CALCULATION

```
[numErrors, ber] = biterr(txBits, rxBits);
```

```
fprintf('Number of bit errors = %d\n', numErrors);  
fprintf('BER = %f\n', ber);
```

```
Number of bit errors = 0  
BER = 0.000000
```

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